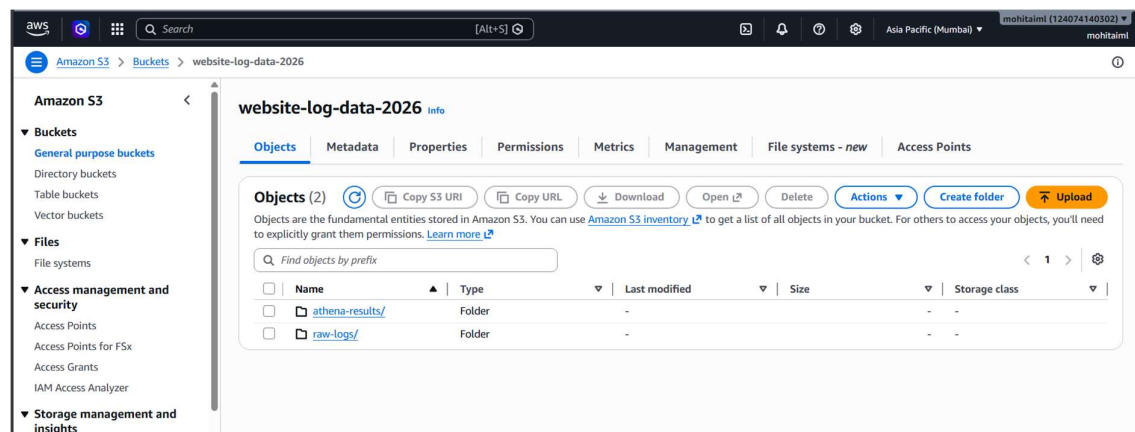


Project: Serverless Website Log Analytics Pipeline

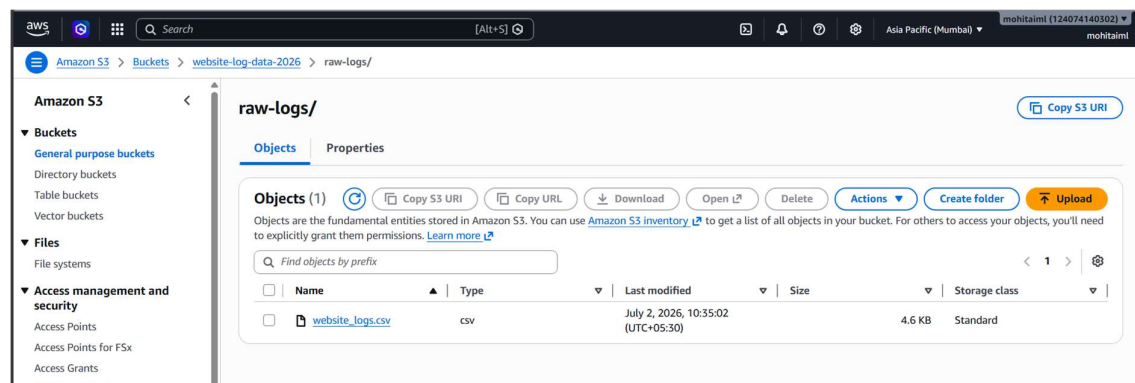
Phase 1: Data Architecture & Storage (Amazon S3)

This phase establishes a clean data lake directory layout where raw logs are separated from automated query outputs.

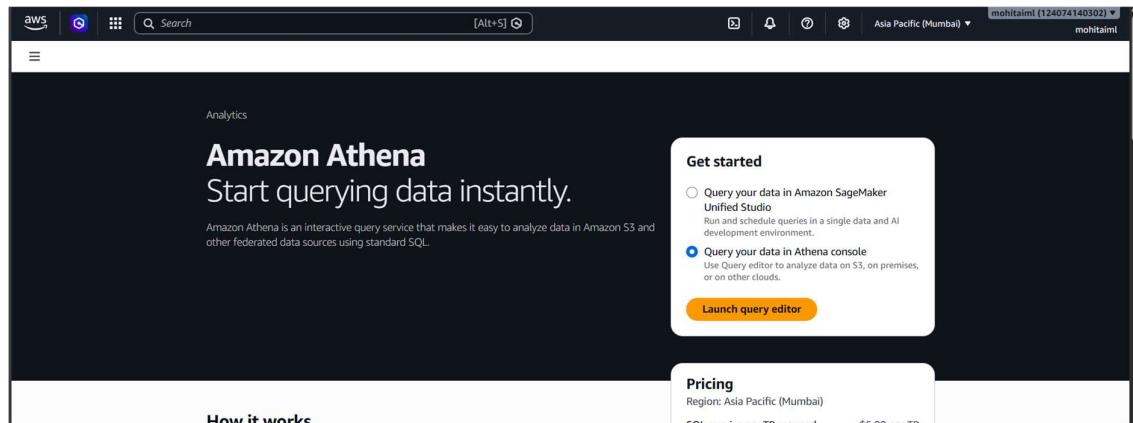
1. **Prepare the Data File:** Create a local file named `website_logs.csv` containing your 100 website traffic records. Ensure the first row contains exactly these headers: `timestamp,user_id,page,country,device`.
2. **Create the S3 Bucket:** Log into the AWS Console, open **Amazon S3**, and click **Create bucket**. Choose a globally unique name (e.g., `website-log-data-2026`).



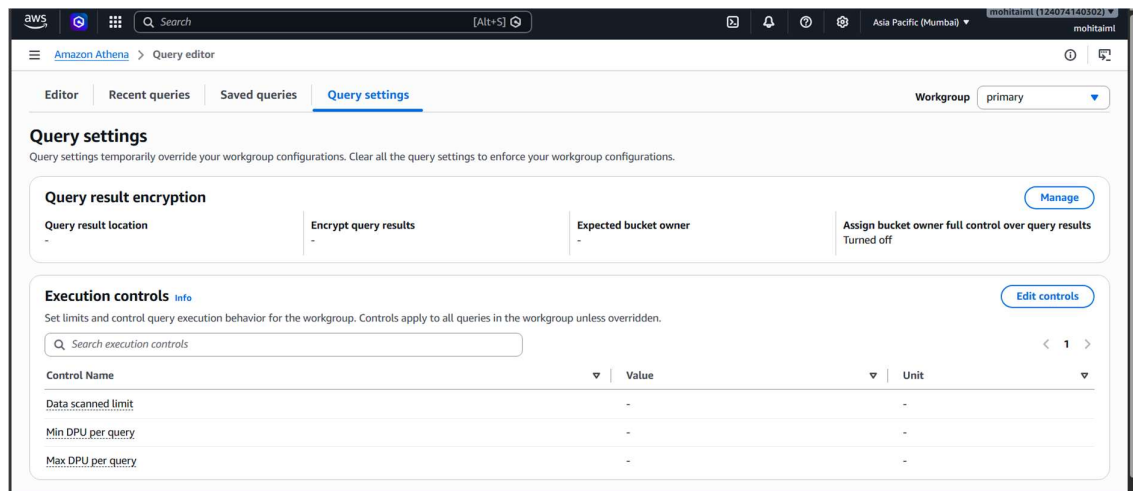
3. **Establish Directory Hierarchy:** Inside your new bucket, create two separate folders to keep your pipeline clean:
 - `raw-logs/` (This will exclusively hold your raw incoming data files).
 - `athena-results/` (This will hold the background CSV files Athena generates during analysis).
4. **Upload Raw Logs:** Navigate inside the `raw-logs/` folder and upload your `website_logs.csv` file there.



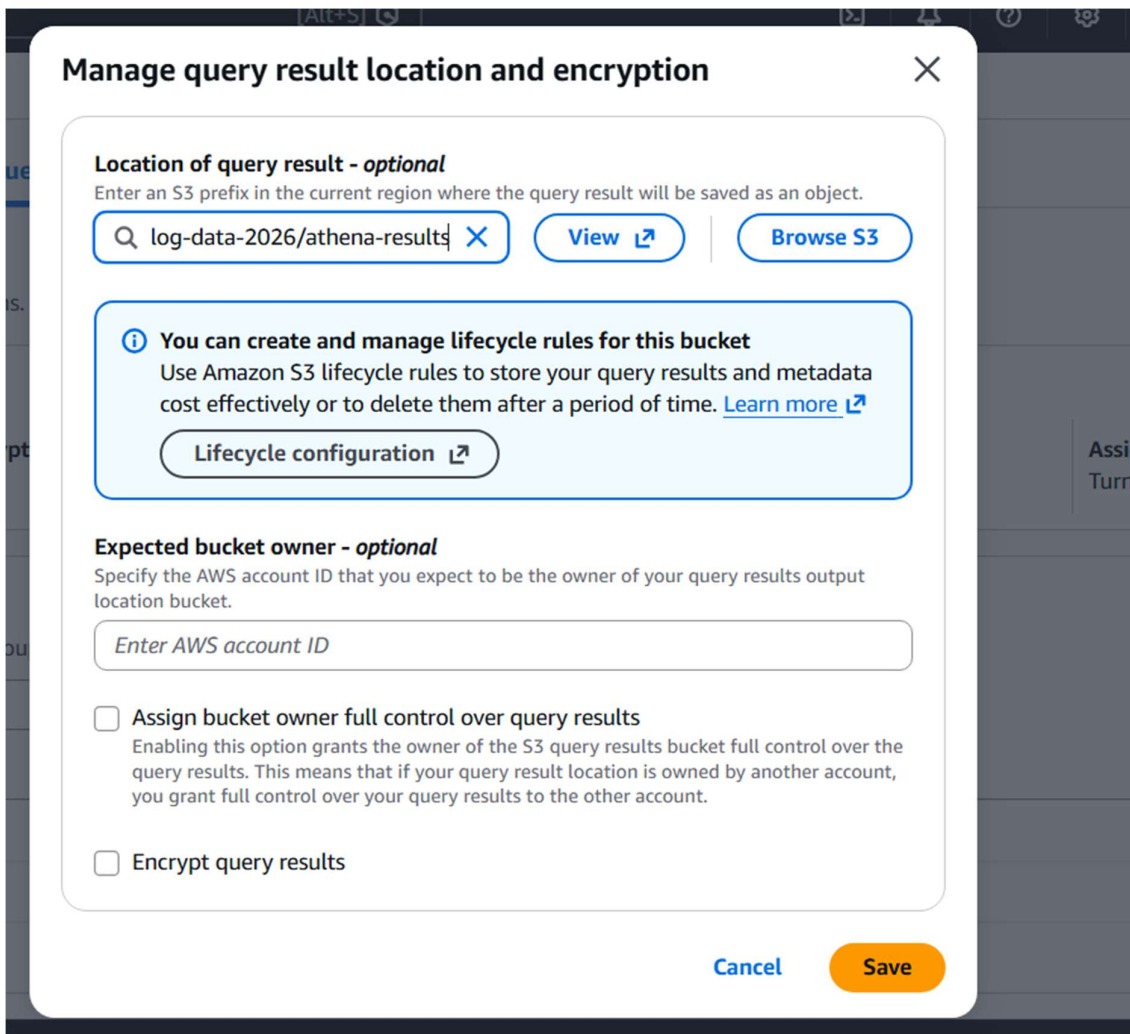
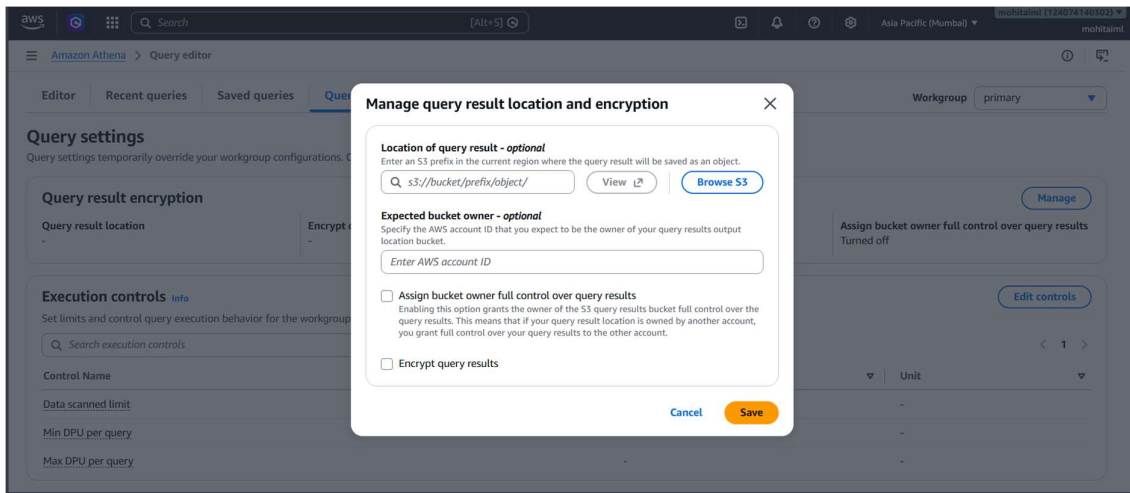
Phase 2: Serverless Database & Schema Definition (Amazon Athena)



This phase configures AWS Glue and Athena to query your folder-isolated data using standard SQL, and validates the data by running your core queries.

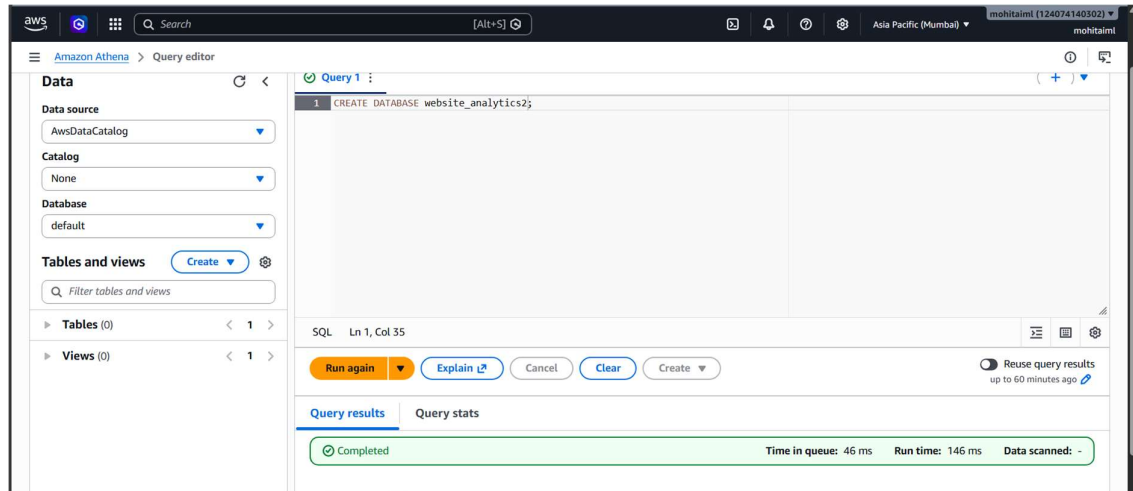


1. **Configure Query Settings:** Open **Amazon Athena** in the AWS Console. Go to the **Settings** tab and click **Manage**. Set the **Query result location** to point directly to your results folder: `s3://website-log-data-2026/athena-results/`. Click **Save**.



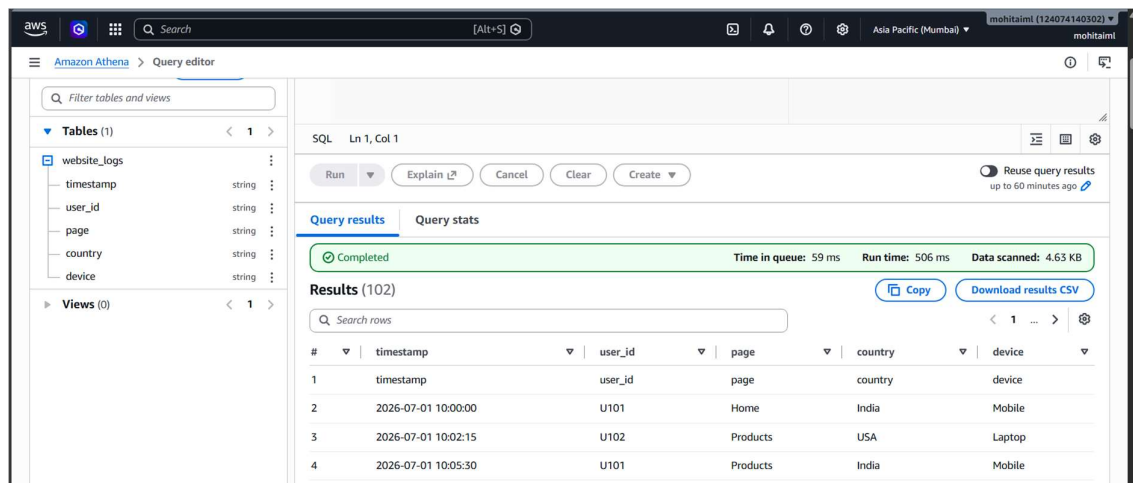
2. **Create the Schema Catalog:** In the query editor, run the following command to spin up your logical database:

CREATE DATABASE website_analytics;



3. **Build the External Table:** Select website_analytics from the Database dropdown menu on the left. Paste and execute the following query. (Note: The LOCATION string points strictly to the raw-logs/ folder so it never reads the results folder recursively.)

```
CREATE EXTERNAL TABLE website_logs (  
    timestamp STRING,  
    user_id STRING,  
    page STRING,  
    country STRING,  
    device STRING  
)  
  
ROW FORMAT DELIMITED  
FIELDS TERMINATED BY ','  
LOCATION 's3://website-log-data-2026/raw-logs/';
```



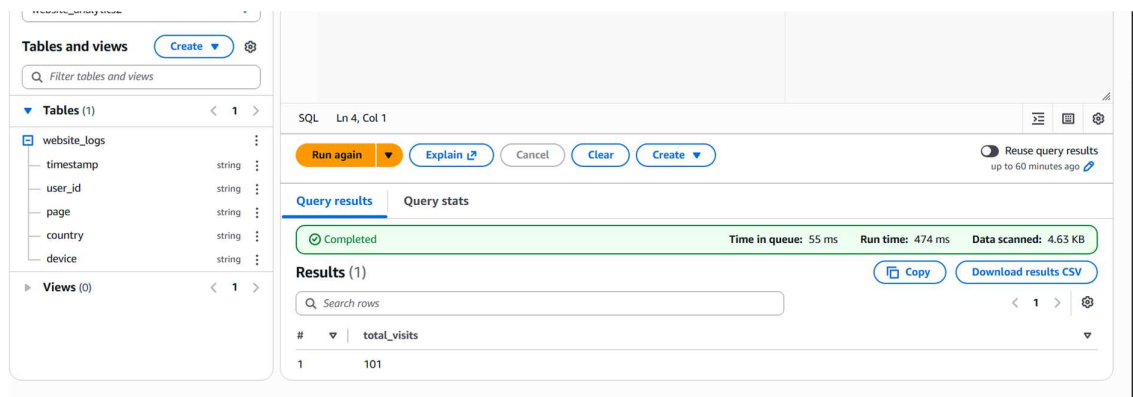
4. **Run Your Core Analytical Queries:** You can run these one by one in Athena to see your aggregated metrics and validate the schema. You will eventually use these exact same datasets in your dashboard.

- **Total Visits:**

SELECT COUNT(*) AS total_visits

FROM website_logs

WHERE timestamp != 'timestamp'; -- This filters out the header row if it exists



- **Visitors by Country:**

SELECT country, COUNT(*) AS visits

FROM website_logs

WHERE country != 'country'

GROUP BY country

ORDER BY visits DESC;

The screenshot shows the Amazon Athena Query Editor interface. On the left, the 'Data' panel shows the 'website_logs' table with columns: timestamp, user_id, page, country, and device. The 'Query 1' editor contains the following SQL:

```
1 SELECT country, COUNT(*) AS visits
2 FROM website_logs
3 WHERE country != 'country'
4 GROUP BY country
5 ORDER BY visits DESC;
```

The query has been executed successfully. The 'Query results' panel shows the following results:

#	country	visits
1	India	22
2	USA	19
3	UK	12

- **Most Visited Pages:**

SELECT page, COUNT(*) AS visits

FROM website_logs

WHERE page != 'page'

GROUP BY page

ORDER BY visits DESC;

The screenshot shows the Amazon Athena Query Editor interface. On the left, the 'Data' panel shows the 'website_logs' table with columns: timestamp, user_id, page, country, and device. The 'Query 1' editor contains the following SQL:

```
1 SELECT page, COUNT(*) AS visits
2 FROM website_logs
3 WHERE page != 'page'
4 GROUP BY page
5 ORDER BY visits DESC;
```

The query has been executed successfully. The 'Query results' panel shows the following results:

#	page	visits
1	Products	28
2	Home	27

Phase 3: Secure Cloud Connectivity (IAM & ODBC DSN)

This phase sets up a secure pipeline allowing your local machine to communicate with the AWS environment without using a root account.

1. Generate Secure IAM Credentials:

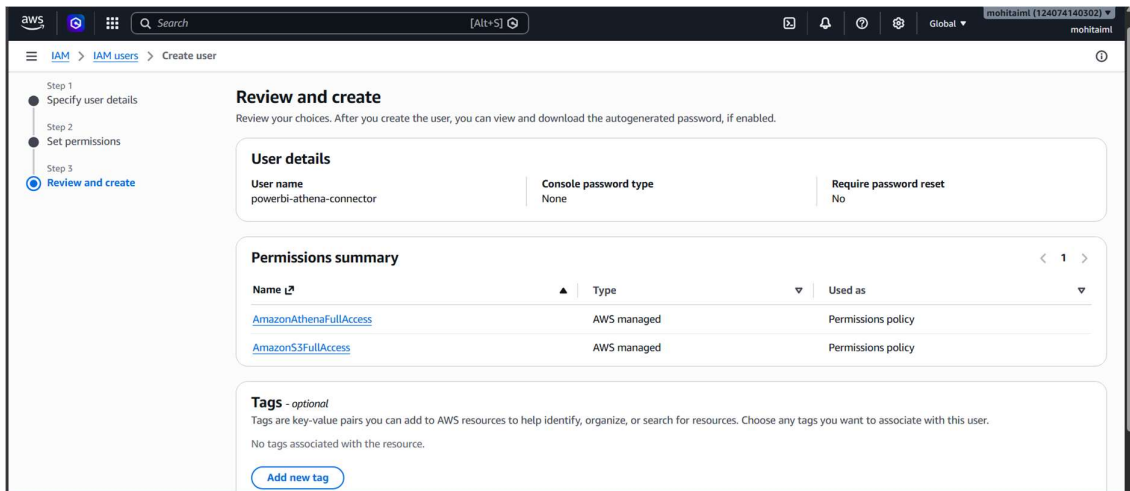
- Open the **AWS IAM Console**.
- Create a dedicated user named powerbi-athena-connector.

The screenshot shows the 'Specify user details' step in the AWS IAM console. The 'User name' field is populated with 'powerbi-athena-connector'. A note below the field states: 'The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and * , . @ _ - (hyphen)'. There is an unchecked checkbox for 'Provide user access to the AWS Management Console - optional'. A blue box contains a tip: 'If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user. [Learn more](#)'. The 'Next' button is highlighted in orange.

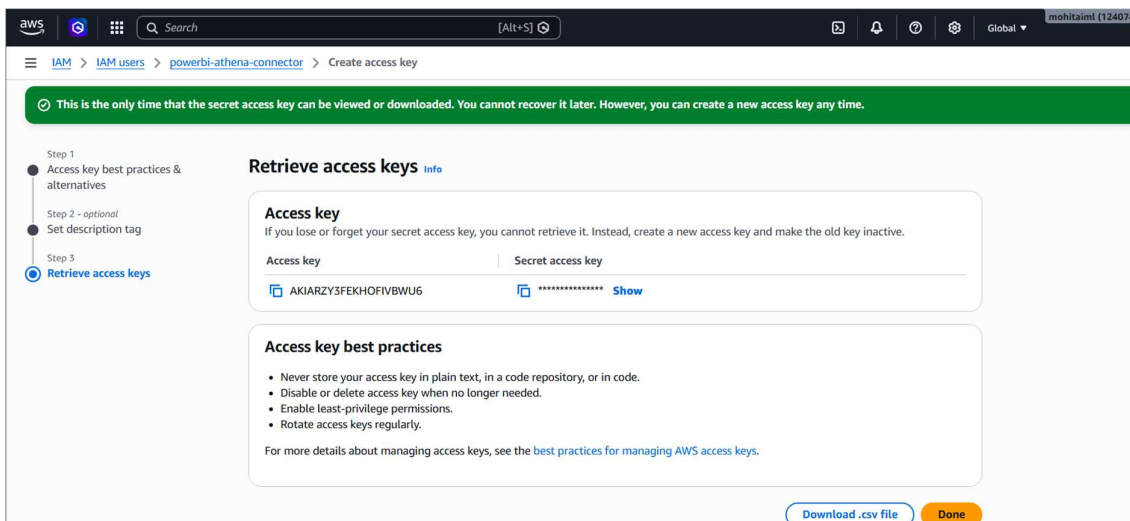
- Attach the AmazonAthenaFullAccess and AmazonS3FullAccess policies directly to this user.

The screenshot shows the 'Set permissions' step in the AWS IAM console. Under 'Permissions options', the 'Attach policies directly' option is selected. Below, the 'Permissions policies (2/1519)' section shows a list of policies. The first two are 'AccessAnalyzerServiceRolePolicy' and 'AccountAccessManagerServiceRolePolicy', both of type 'AWS managed' and with 0 attached entities.

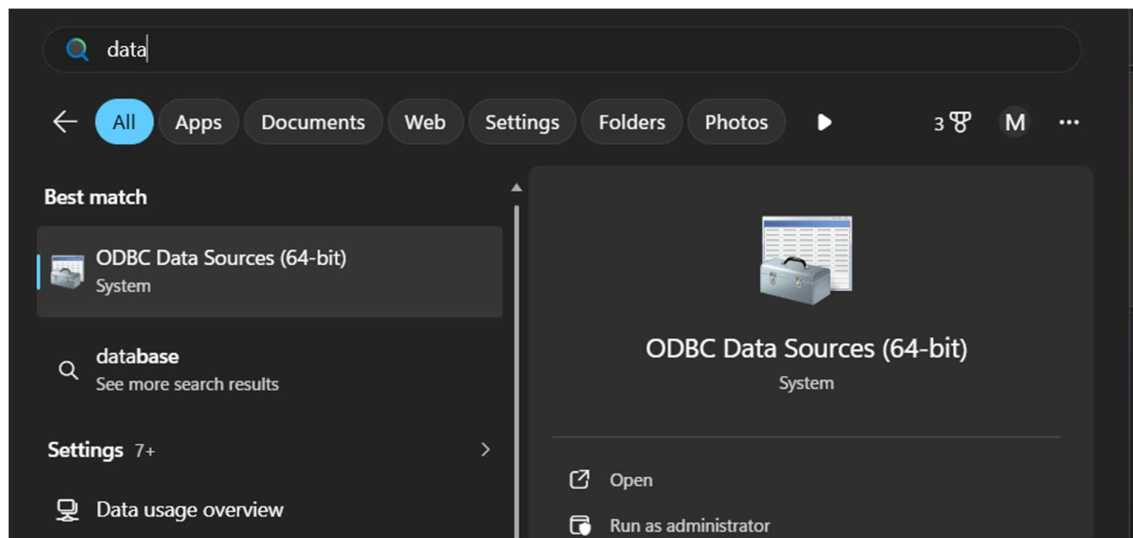
Policy name	Type	Attached entities
AccessAnalyzerServiceRolePolicy	AWS managed	0
AccountAccessManagerServiceRolePolicy	AWS managed	0



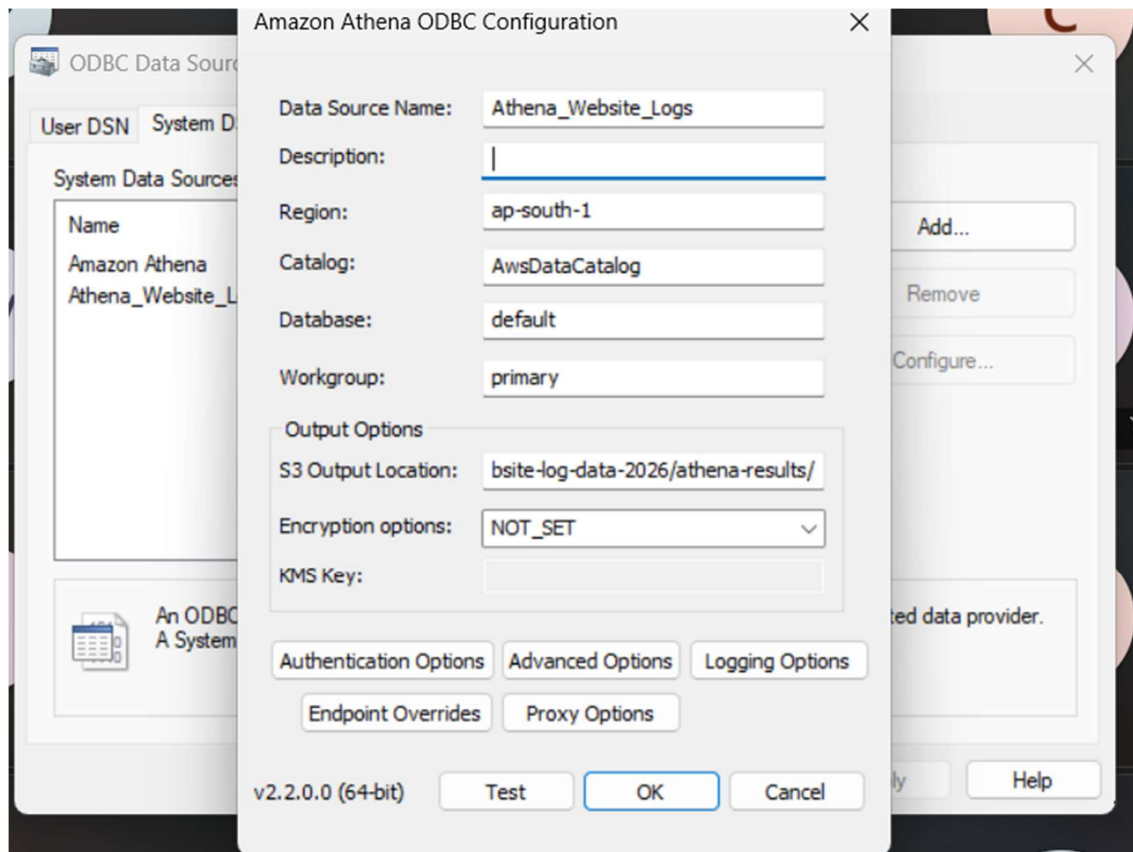
- Go to the user's **Security credentials** tab, click **Create access key**, choose **Third-party service**, and copy your new **Access Key ID** and **Secret Access Key**.



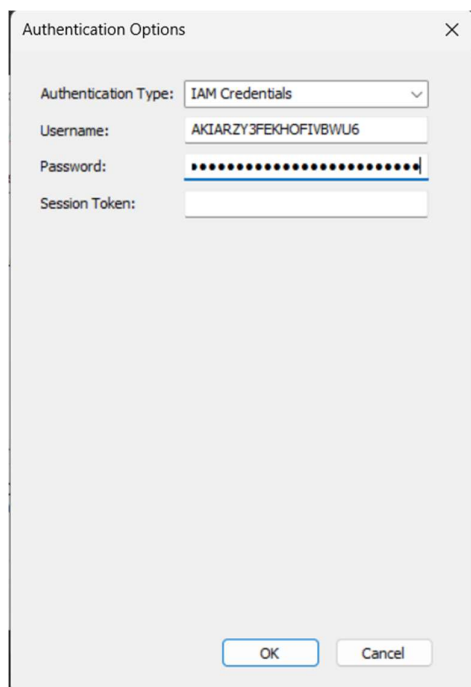
2. **Install Driver Software:** Download and run the official **Simba Athena 64-bit ODBC Driver** installer on your Windows machine.
3. **Configure the System DSN:**
 - Search your computer for **ODBC Data Sources (64-bit)** and run it.



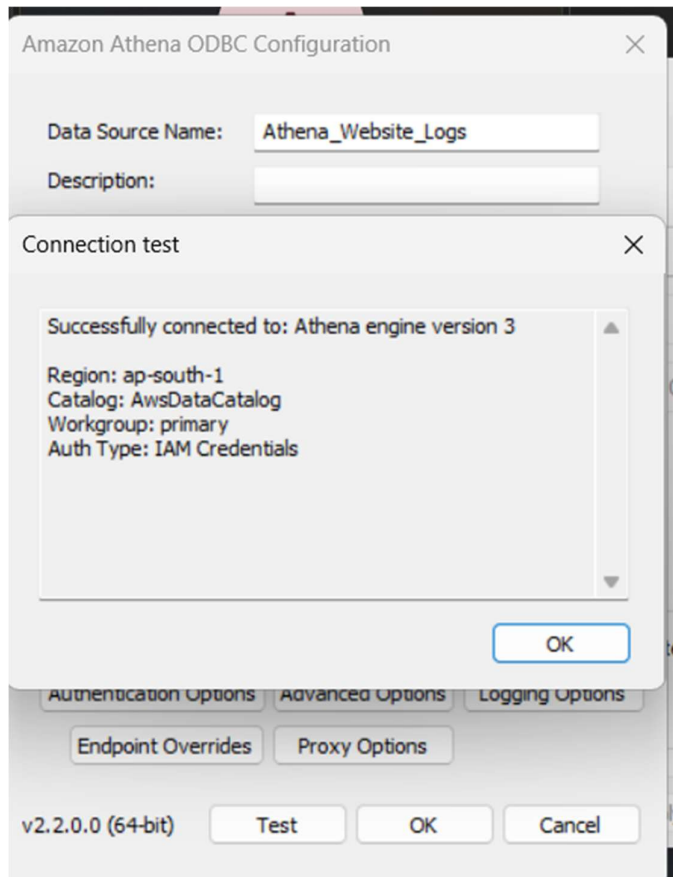
- Navigate to the **System DSN** tab and click **Add....**
- Select the **Simba Athena ODBC Driver** from the list and click **Finish**.
- Fill out the configuration window:
 - **Data Source Name:** Athena_Website_Logs
 - **AWS Region:** Your bucket's region (e.g., us-east-1)
 - **S3 Output Location:** s3://website-log-data-2026/athena-results/



- Click **Authentication Options**, select **IAM Credentials**, paste your Access Key into the **User** field, paste your Secret Key into the **Password** field, and click **OK**.

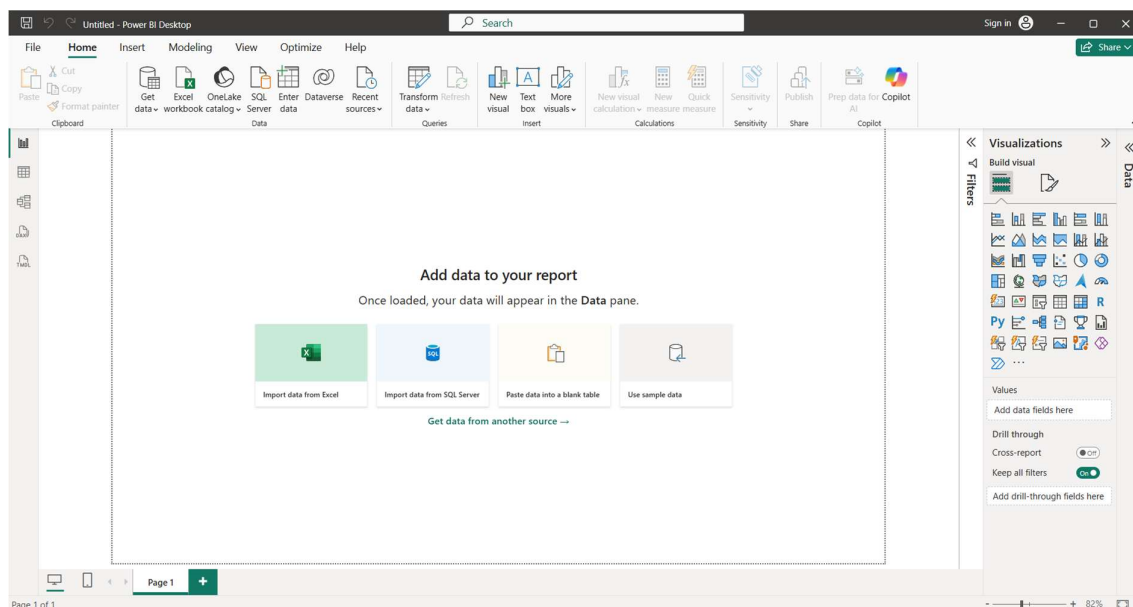


- Click **Test** to ensure a successful cloud connection, then save it.



Phase 4: BI Dashboard Visualization (Power BI Desktop)

This phase imports the live cloud schema, transforms the data to remove CSV structural text, and generates your executive dashboard.



From ODBC

Data source name (DSN)

Amazon Athena

Advanced options

OK

Cancel

1. **Ingest the Athena Schema:** Open **Power BI Desktop**. Click **Get Data > More... > ODBC**. Select **Athena_Website_Logs** from the dropdown and hit **OK**.
2. **Load into Power Query:** In the Navigator pop-up, expand **AwsDataCatalog > website_analytics**, check the box next to your **website_logs** table, and click **Transform Data**.

The screenshot shows the Power BI Navigator window. On the left, the 'Navigator' pane displays a tree view of data sources. Under 'AwsDataCatalog [2]', the 'website_analytics' folder is expanded, and the 'website_logs' table is selected with a checkmark. On the right, a preview of the 'website_logs' table is shown with columns: timestamp, user_id, page, country, and device. The table contains 15 rows of data. At the bottom of the window, there are three buttons: 'Load', 'Transform Data', and 'Cancel'.

timestamp	user_id	page	country	device
2026-07-01 10:00:00	U101	Home	India	Mobile
2026-07-01 10:02:15	U102	Products	USA	Laptop
2026-07-01 10:05:30	U101	Products	India	Mobile
2026-07-01 10:07:45	U103	Home	UK	Desktop
2026-07-01 10:10:00	U104	Blog	Canada	Tablet
2026-07-01 10:12:10	U105	Home	Australia	Mobile
2026-07-01 10:14:25	U102	Checkout	USA	Laptop
2026-07-01 10:15:40	U106	Products	Germany	Desktop
2026-07-01 10:18:55	U103	Contact	UK	Desktop
2026-07-01 10:20:10	U107	Home	India	Mobile
2026-07-01 10:22:25	U108	Blog	USA	Mobile
2026-07-01 10:25:30	U109	Products	Brazil	Laptop
2026-07-01 10:27:45	U107	Products	India	Mobile
2026-07-01 10:30:00	U110	Home	Japan	Tablet
2026-07-01 10:32:15	U101	Checkout	India	Mobile
2026-07-01 10:34:20	U111	Home	France	Desktop
2026-07-01 10:35:45	U106	Pricing	Germany	Desktop
2026-07-01 10:38:10	U112	Products	USA	Laptop
2026-07-01 10:40:25	U109	Checkout	Brazil	Laptop
2026-07-01 10:42:40	U113	Home	India	Mobile
2026-07-01 10:45:00	U114	Contact	Canada	Mobile
2026-07-01 10:47:15	U110	Products	Japan	Tablet
2026-07-01 10:50:30	U115	Home	UK	Laptop

3. Data Transformation & Cleaning:

- **Filter out CSV Headers:** Look at your rows. Click the small downward-pointing filter arrow on the right side of the timestamp column header. Scroll down the checklist, **uncheck** the literal text value "timestamp", and click **OK** to remove the header row.
 - **Optimize Schema:** Right-click the timestamp column header and select **Remove** to save storage space and memory.
 - **Apply Changes:** Click the large **Close & Apply** icon in the upper-left corner of the window.
4. **Build Visual components:** Look at the **Data** pane on the far-right side of your screen. Drag and drop your fields onto the empty white canvas using these configurations:
- **Total Visits Metric:** Add a **Card** visual (the 123 icon). Drag user_id into the Fields well. Right-click it inside the well and set it to **Count**. It will cleanly display your total volume.
 - **Geographic Traffic Distribution:** Add a **Stacked Bar Chart**. Drag country to the **Y-axis** and user_id to the **X-axis** (set aggregation to Count).
 - **Platform Split:** Add a **Pie Chart**. Drag device to the **Legend** and user_id to the **Values** box (set aggregation to Count).
 - **Page Engagement Performance:** Add a **Stacked Column Chart**. Drag page to the **X-axis** and user_id to the **Y-axis** (set aggregation to Count).

